

**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING**

Khwopa College Of Engineering
Libali, Bhaktapur

Department of Computer and Electronics Engineering



**A PROPOSAL ON
AI-BASED NEPALI PRONUNCIATION
EVALUATION SYSTEM**

Submitted in partial fulfillment of the requirements for the degree
Bachelor of Computer Engineering

Submitted by

Rakshyak Pradhananga	KCE080BCT027
Ritesh Shahi	KCE080BCT032
Shreya Khatri	KCE080BCT038
Suvanjan Dahal	KCE080BCT044

Khwopa College of Engineering
Libali, Bhaktapur

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Head of Department
Department of Electronics and Computer Engineering
Khwopa College of Engineering
Libali, Bhaktapur, Nepal

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Rakshyak Pradhananga	KCE080BCT027
Ritesh Shahi	KCE080BCT032
Shreya Khatri	KCE080BCT038
Suvanjan Dahal	KCE080BCT044

Abstract

This project presents an AI-Based Nepali Pronunciation Evaluation System designed to assist learners and speakers in assessing the quality of their Nepali pronunciation. The system evaluates a user's spoken utterance by comparing it with a standard reference pronunciation and provides an overall pronunciation score along with a quality classification.

The proposed system follows a speech processing and machine learning pipeline. A Nepali sentence is displayed to the user, who then records their pronunciation. The recorded speech undergoes preprocessing and feature extraction using Mel Frequency Cepstral Coefficients (MFCC), which capture the important acoustic characteristics of speech. Dynamic Time Warping (DTW) is employed to measure the similarity between the user's speech and the reference speech while accommodating variations in speaking rate and timing.

The extracted acoustic features and similarity measures are combined to form a feature vector that is used to train and evaluate a Support Vector Machine (SVM) classifier. The classifier predicts the overall pronunciation quality and categorizes the pronunciation into predefined classes such as Good, Average, or Poor. In addition, a pronunciation score is generated to provide quantitative feedback to the user.

Unlike advanced pronunciation assessment systems that perform phoneme-level error detection and correction, the proposed system focuses on sentence-level pronunciation evaluation. Therefore, it provides an overall assessment of pronunciation quality rather than identifying specific phoneme-level mispronunciations. The system offers a simple, accessible, and automated approach for Nepali pronunciation evaluation and can serve as a foundation for future research in computer-assisted pronunciation assessment for the Nepali language.

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List of Symbols and Abbreviations

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CAPT	Computer-Assisted Pronunciation Training
DTW	Dynamic Time Warping
FFT	Fast Fourier Transform
MFCC	Mel-Frequency Cepstral Coefficients
NLP	Natural Language Processing
SVM	Support Vector Machine
TU	Tribhuvan University

Chapter 1

Introduction

1.1 Background Introduction

Clear pronunciation is essential for effective spoken communication. For Nepali speakers, improving pronunciation can be challenging due to limited access to proper guidance and practice tools. Traditional learning methods rely on human instructors, which may not always be available, consistent, or affordable.

To address this challenge, automated systems can be used to evaluate pronunciation and provide immediate feedback. Computer-Assisted Pronunciation Training (CAPT) systems utilize speech processing techniques to analyze and compare user speech with standard reference pronunciations.

In this project, efficient and widely used methods are adopted, including MFCC for extracting speech features, DTW for measuring similarity between user and reference speech, and SVM for classification. The system evaluates pronunciation and classifies it into three categories: Good, Fair, and Poor, offering a simple and accessible solution for Nepali pronunciation assessment.

1.2 Motivation

Most existing speech systems focus on recognizing spoken content rather than evaluating pronunciation quality. Additionally, many commercial solutions are expensive and not tailored to Nepali users, limiting their usability.

This project is motivated by the need for a simple, affordable, and user-friendly system that can provide quick pronunciation evaluation. By offering clear results in the form of Good, Fair, or Poor, the system enables users to easily understand their performance and practice effectively.

1.3 Problem Definition

Accurate and consistent pronunciation evaluation tools for Nepali learners are limited. While existing technologies can recognize speech, they often do not provide meaningful assessment of pronunciation quality.

This project addresses the problem by developing a method to compare user speech with reference audio and categorize pronunciation into clear and interpretable quality levels.

1.4 Goals and Objectives

Goal: Develop a simple AI-based system for evaluating Nepali pronunciation and classifying it into Good, Fair, and Poor categories.

Objectives:

1. Extract acoustic features from speech using MFCC.
2. Measure similarity between user and reference speech using Dynamic Time Warping (DTW).
3. Generate training labels based on DTW distance thresholds.
4. Train an SVM model to classify pronunciation quality.
5. Develop a lightweight user interface for recording speech and displaying results.

1.5 Scope and Applications

Scope:

- The system focuses on basic pronunciation evaluation for Nepali speakers.
- It processes short words or sentences as input.
- Output is limited to simple classification: Good, Fair, or Poor.
- The system runs locally using a lightweight interface such as Streamlit.

Applications:

- Self-learning tool for students to practice and monitor pronunciation independently.
- Supporting tool in schools and colleges for basic speaking practice.
- Can be integrated into e-learning platforms for automated assessment.
- Useful for checking pronunciation before presentations, interviews, or public speaking.
- Acts as a quick screening tool to evaluate speaking quality in training sessions.
- Can assist beginners in language learning by providing simple performance feedback.

Chapter 2

Literature Review

This chapter reviews previous research related to automatic pronunciation evaluation, speech feature extraction, acoustic similarity measurement, and machine learning techniques used in speech processing systems. The reviewed works provide the theoretical foundation for developing an automatic Nepali pronunciation evaluation system using MFCC, DTW, and SVM techniques.

2.1 MFCC for Speech Representation

Mel-Frequency Cepstral Coefficients (MFCC) are one of the most commonly used feature extraction techniques in speech processing. MFCC converts speech signals into a set of numerical features that represent important spectral characteristics of human speech. Since MFCC is based on the Mel scale, it provides a representation closer to human auditory perception.

Davis and Mermelstein (1980) introduced MFCC-based feature extraction and demonstrated its effectiveness in speech recognition applications. Their work showed that MFCC provides better representation of speech characteristics compared with traditional spectral analysis methods. MFCC has since become a standard feature extraction method in speech recognition, speaker identification, and pronunciation analysis.

2.2 DTW for Pronunciation Similarity

Dynamic Time Warping (DTW) is a time-series alignment algorithm widely used for comparing speech signals with different speaking rates. In pronunciation evaluation, speakers may pronounce the same word or sentence with different durations. DTW addresses this by finding the optimal alignment path between two speech feature sequences.

Sakoe and Chiba (1978) introduced DTW optimization techniques for speech recognition and demonstrated its ability to handle temporal variations in speech. When combined with MFCC features, DTW can calculate the acoustic distance between reference pronunciation and learner pronunciation.

2.3 Machine Learning for Pronunciation Classification

Machine learning approaches are widely applied in speech analysis to improve classification performance. Unlike threshold-based methods, machine learning models can learn patterns from extracted speech features and provide more reliable results.

Support Vector Machine (SVM) is a supervised learning algorithm commonly used for speech classification due to its effectiveness with high-dimensional data. Muda et al. (2010) investigated speech recognition using MFCC, DTW, and classification techniques, showing that combining feature extraction with machine learning improves performance compared with conventional approaches.

2.4 Automatic Pronunciation Assessment Systems

Automatic pronunciation assessment is an important area in Computer-Assisted Pronunciation Training (CAPT) systems. These systems aim to provide learners with automatic feedback without requiring continuous evaluation from human instructors.

Neri et al. (2002) studied the use of speech recognition technology for second language learning and demonstrated that computer-based systems can analyze learner speech and provide pronunciation feedback.

2.5 Acoustic Feature-Based Pronunciation Analysis

Pronunciation quality can be evaluated by analyzing acoustic differences between correct and incorrect speech. Acoustic feature-based methods compare numerical characteristics extracted directly from speech signals rather than relying on complete phonetic transcription.

Franco et al. (1997) investigated acoustic-based pronunciation evaluation methods and showed that acoustic information can represent pronunciation differences, highlighting the importance of speech features in detecting variation in learner speech.

2.6 Pronunciation Analysis in South Asian Context

Speech processing techniques have been applied to several South Asian languages, where speech resources are often limited. Studies suggest that approaches such as MFCC and DTW can still provide useful results with smaller datasets.

Since Nepali is a low-resource language compared with widely studied languages such as English, similar acoustic-based approaches can be applied. The availability of datasets such as OpenSLR143 provides opportunities for Nepali speech analysis.

2.7 Acoustic Distance-Based Evaluation

Acoustic distance-based evaluation measures pronunciation similarity by calculating differences between speech feature representations. These methods provide numerical similarity scores that can be used for evaluation or classification.

Bartelds et al. (2020) proposed an acoustic distance measurement approach using MFCC features and DTW alignment. Their results indicated that acoustic distance values correlate with human pronunciation judgments.

Author/Year	Method	Dataset	Key Findings	Limitations
Davis & Mermelstein (1980)	MFCC Features	Speech datasets	Effective speech representation	Sensitive to noise
Sakoe & Chiba (1978)	DTW Algorithm	Speech signals	Handles timing variation	High computation cost
Muda et al. (2010)	MFCC + DTW + SVM	Speech datasets	Improved classification accuracy	Requires parameter tuning
Franco et al. (1997)	Acoustic Features	Learner speech	Detected pronunciation errors	Limited feature scope
Neri et al. (2002)	CAPT System	Learner speech	Automatic feedback generation	Requires trained speech models
Bartelds et al. (2020)	MFCC + DTW Distance	Speech corpus	Correlates with human judgment	Affected by recording conditions

Table 2.1: Review Matrix of Related Works

From the reviewed studies, it can be concluded that MFCC provides effective speech feature representation, while DTW enables similarity measurement between speech signals. Machine learning models such as SVM further enhance classification by learning patterns from acoustic features.

Based on these findings, this project combines MFCC for speech feature extraction, DTW for pronunciation similarity measurement, and SVM for classification to develop an automatic Nepali pronunciation evaluation system using the OpenSLR143 dataset.

Chapter 3

Requirement Analysis

3.1 Software Requirements

- **Operating System:** Ubuntu 20.04 / Windows 10 or later
- **Programming Language:** Python 3.9+
- **Speech Processing:** Librosa, NumPy, SciPy
- **Feature Extraction:** Mel Frequency Cepstral Coefficients (MFCC)
- **Similarity Measurement:** Dynamic Time Warping (DTW)
- **Machine Learning:** Scikit-learn (Support Vector Machine - SVM)
- **Dataset:** Nepali Speech Dataset (e.g., OpenSLR Nepali Corpus)
- **UI Framework:** Streamlit
- **Audio Handling:** SoundDevice, SoundFile, PyAudio
- **Version Control:** Git / GitHub

3.2 Hardware Requirements

- **Minimum RAM:** 8 GB
- **Storage:** 5 GB free space
- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher
- **GPU (Optional):** Not required
- **Microphone:** Standard USB or built-in microphone (16 kHz capable)

3.3 Functional Requirements

- The system shall display a target Nepali sentence to the user.
- The system shall allow the user to record their pronunciation of the displayed sentence.
- The system shall preprocess the recorded speech to remove noise and prepare it for analysis.
- The system shall extract acoustic features from the recorded speech using MFCC.
- The system shall compare the user's speech with a reference pronunciation using DTW.
- The system shall create a feature vector using MFCC features and DTW similarity measures.
- The system shall classify pronunciation quality using a trained Support Vector Machine (SVM) model.
- The system shall provide an overall pronunciation score.
- The system shall categorize pronunciation quality as Good, Average, or Poor.
- The system shall display basic speech visualizations such as waveform, spectrogram, and MFCC plots.
- The system shall store and process reference audio-sentence pairs from the Nepali speech dataset.

3.4 Non-Functional Requirements

3.4.1 Reliability

The system shall produce consistent pronunciation scores and classifications for identical audio inputs under similar recording conditions.

3.4.2 Performance

The system shall complete the pronunciation evaluation process within 5 seconds on the minimum hardware specification after the audio recording is finished.

3.4.3 Accuracy

The SVM classifier shall achieve satisfactory classification performance on the selected Nepali speech dataset. Performance shall be evaluated using standard metrics such as Accuracy, Precision, Recall, and F1-Score.

3.4.4 Robustness

The system shall tolerate moderate background noise and recording variations without crashing. If the audio quality is insufficient, an appropriate error message shall be displayed to the user.

3.5 Feasibility Study

3.5.1 Time Feasibility

The project is designed to be completed within one academic semester (approximately 16 weeks). The use of established speech processing techniques such as MFCC and DTW, along with a traditional machine learning classifier (SVM), reduces implementation complexity and makes the project achievable within the available timeframe.

3.5.2 Economic Feasibility

The project relies entirely on open-source software and publicly available datasets. Python, Librosa, Scikit-learn, Streamlit, and the selected Nepali speech dataset are freely available, eliminating licensing costs and minimizing development expenses.

3.5.3 Technical Feasibility

The project utilizes well-established speech processing and machine learning techniques. The development team possesses the necessary background in Python programming, signal processing, and machine learning acquired through academic coursework and practical projects.

3.5.4 Operational Feasibility

The system provides a simple and user-friendly interface through Streamlit. Users only need a computer and a microphone to perform pronunciation evaluation. The system can operate on standard personal computers without requiring specialized hardware.

3.5.5 Project Limitation

The proposed system performs sentence-level pronunciation evaluation and provides an overall pronunciation score and quality classification. It does not perform phoneme-level mispronunciation detection, phoneme segmentation, or detailed articulatory feedback. Therefore, the system can indicate the overall quality of pronunciation but cannot identify the specific phonemes or words responsible for pronunciation errors.

Chapter 4

Methodology

4.1 Software Development Approach

The project follows an iterative Software Development Life Cycle (SDLC) approach to develop an AI-based Nepali pronunciation evaluation system. The system is implemented in a modular structure consisting of data handling, speech processing, feature extraction, similarity measurement, and classification. The OpenSLR143 Nepali speech dataset is used for training and evaluation. Python is used as the main programming language, with Librosa for MFCC extraction and audio processing. Dynamic Time Warping (DTW) is applied to measure similarity between reference and user speech signals. The extracted MFCC features and DTW distance values are combined to form feature vectors, which are used to train a Support Vector Machine (SVM) classifier that categorizes pronunciation quality into *Good*, *Average*, and *Poor*. Streamlit is used to build the user interface for recording speech and displaying results, and the system is developed iteratively to improve accuracy and performance.

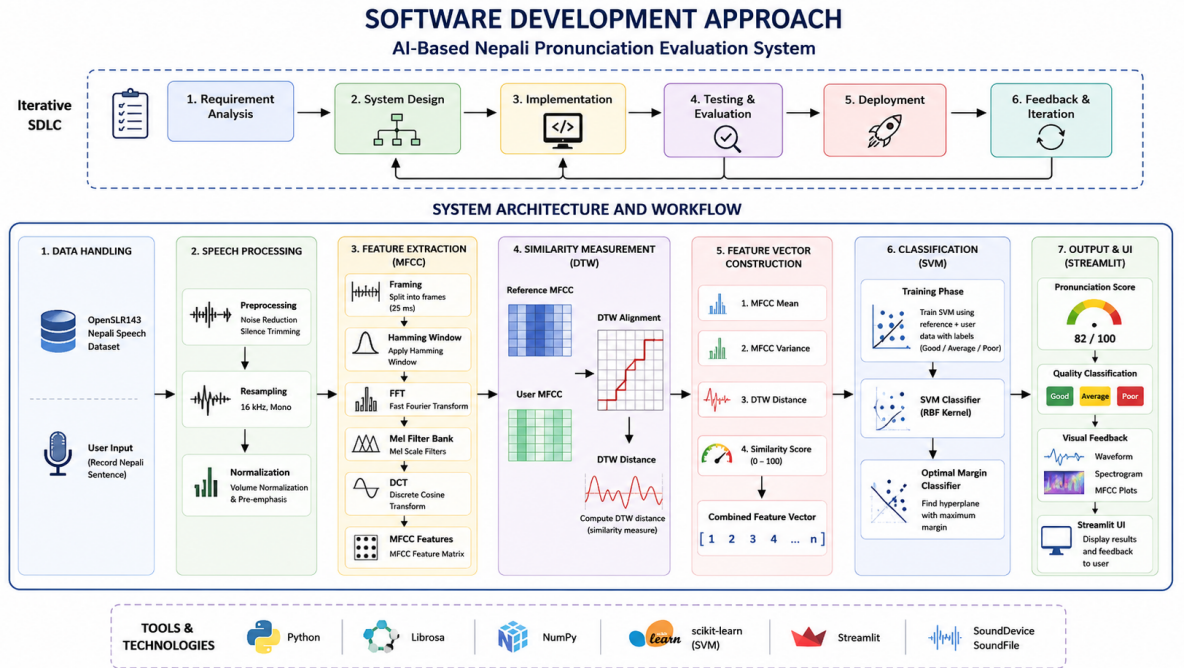


Figure 4.1: Software Development Approach

4.2 System Design and Architecture

4.2.1 Block Diagram

The overall system architecture consists of a speech processing pipeline followed by a machine learning-based pronunciation evaluation module. The user records a Nepali sentence, which is then processed and compared with a reference pronunciation. The resulting features are classified to generate a pronunciation score and quality label

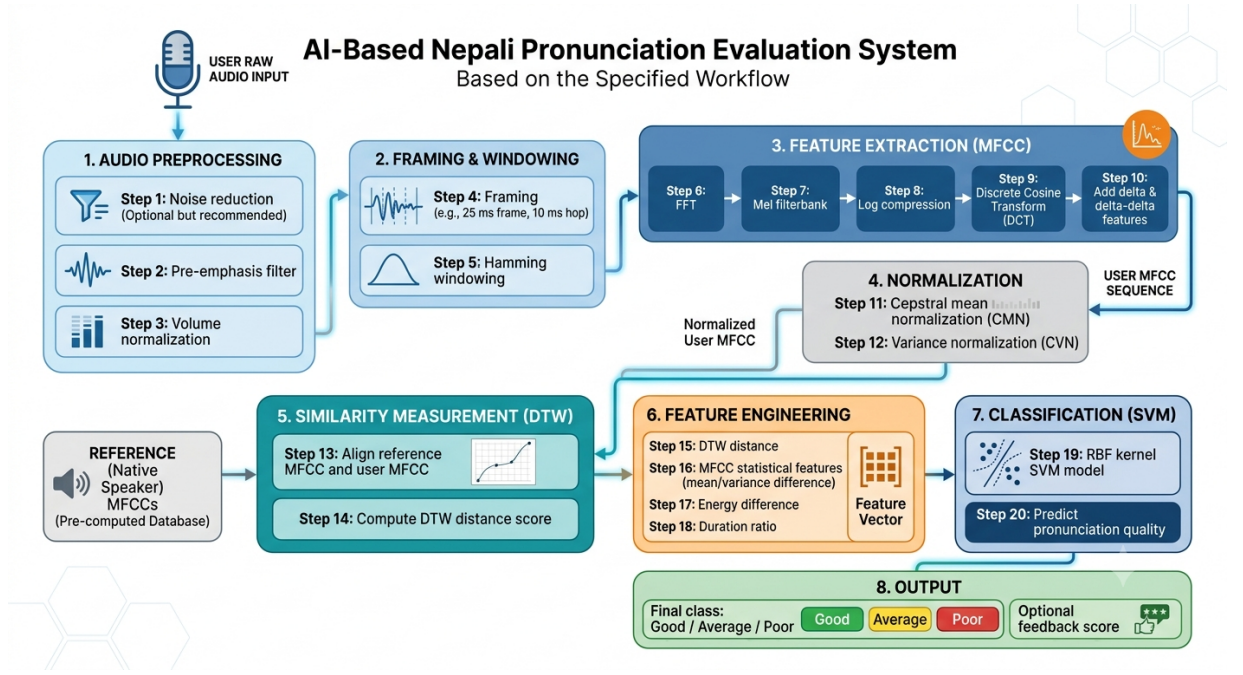


Figure 4.2: Block-Diagram of Nepali Pronunciation Evaluation System

4.2.2 Use Case Diagram

The user can select a Nepali sentence, record speech, submit the recording for evaluation, and receive an overall pronunciation score and quality classification.

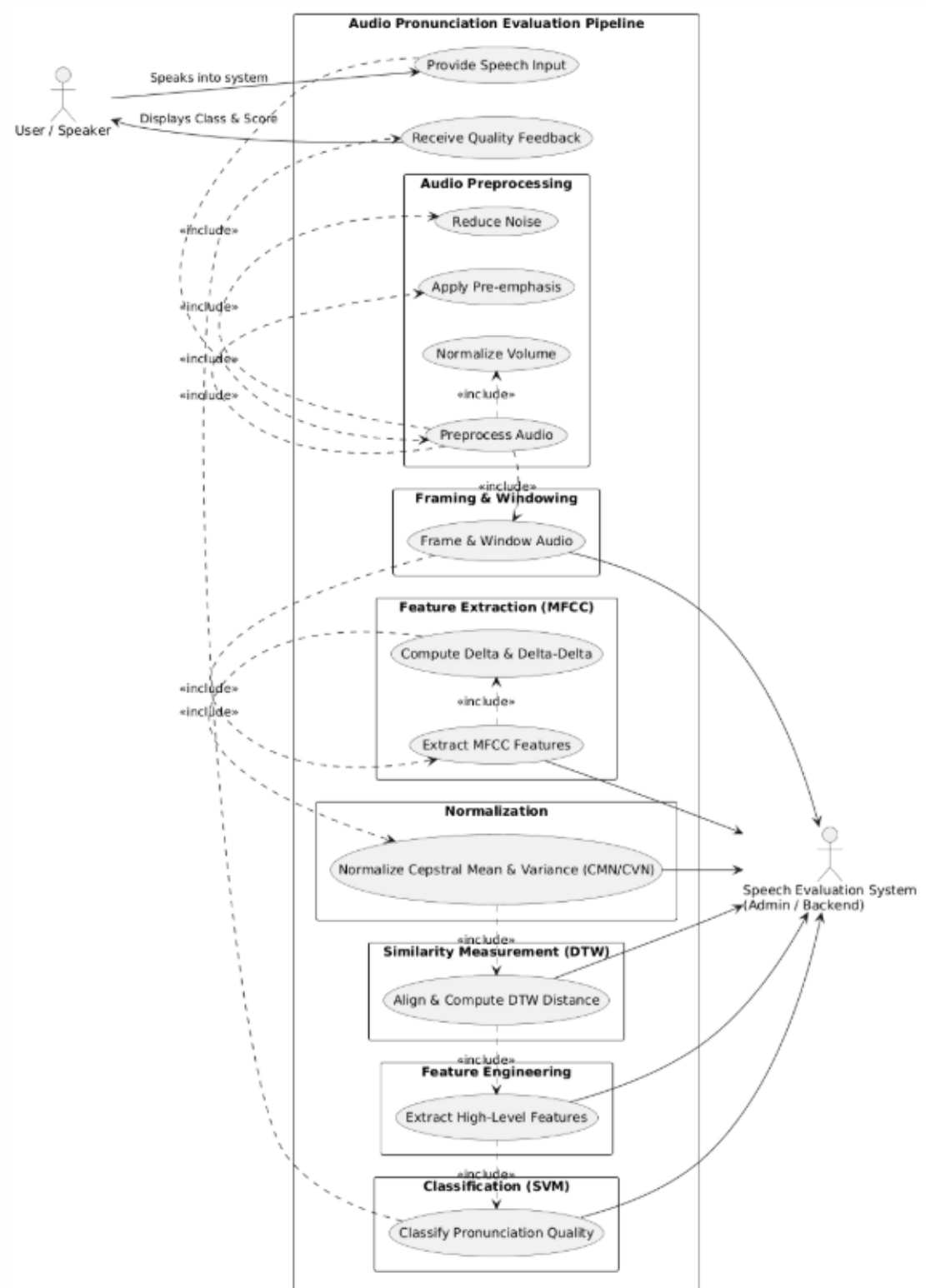


Figure 4.3: Use Case Diagram of Nepali Pronunciation Evaluation System

4.3 Operational Workflow of the System

4.3.1 Data Collection

The system uses a publicly available Nepali speech dataset, specifically the OpenSLR Nepali corpus, for training and evaluation. This dataset includes speech recordings along with their corresponding text transcripts. From this dataset, selected audio-text pairs are used as reference pronunciations, which serve as standard samples for comparison during evaluation.

4.3.2 Data Preprocessing

All audio recordings are first standardized by resampling them to 16 kHz and converting them to mono format. Basic preprocessing steps such as silence removal, noise reduction, and amplitude normalization are applied to improve audio quality and maintain consistency across samples. After preprocessing, Mel Frequency Cepstral Coefficients (MFCC) are extracted from both the reference and user recordings. Dynamic Time Warping (DTW) is then used to align the speech sequences, allowing comparison even when there are differences in speaking speed.

4.3.3 Feature Extraction

MFCC features are used to capture key acoustic properties of speech. In addition to MFCCs, DTW distance values are calculated to measure similarity between the user's pronunciation and the reference. Statistical features such as mean and variance of MFCC coefficients are also included. These components are combined to form a feature vector representing pronunciation characteristics.

4.3.4 Model Selection

A Support Vector Machine (SVM) classifier is used to categorize pronunciation into three classes: *Good*, *Average*, and *Poor*. The model is trained using feature vectors derived from MFCC and DTW-based measurements. SVM is chosen because it performs well with high-dimensional data and is effective even with moderate-sized datasets.

4.3.5 Model Evaluation Metrics

To evaluate the system, standard performance metrics are used, including Accuracy, Precision, Recall, and F1-Score. Accuracy indicates the overall correctness of predictions, while Precision and Recall provide more detailed insights into classification performance. The F1-Score provides a balanced measure of Precision and Recall, giving a more comprehensive evaluation.

4.3.6 Pronunciation Evaluation Process

In the evaluation stage, the system displays a Nepali sentence to the user, who records their pronunciation. The recorded audio is then preprocessed and converted into MFCC features. DTW is used to compute similarity between the

user’s speech and the reference pronunciation. A feature vector is constructed using MFCC statistics and DTW distance. This feature vector is then passed to a trained Support Vector Machine (SVM) model, which predicts the pronunciation quality as *Good*, *Average*, or *Poor*.

4.4 System Limitation

The proposed system has the following limitations:

- The system provides only sentence-level pronunciation evaluation and does not perform phoneme-level analysis.
- It cannot identify specific mispronounced phonemes or words responsible for pronunciation errors.
- The system does not provide detailed articulatory feedback such as tongue position, stress, or intonation correction.
- The model is trained on a limited Nepali dataset (OpenSLR143), which may affect its ability to generalize across different speakers and accents.
- The system evaluates only predefined sentences and is not designed for free-form or spontaneous speech input.

Chapter 5

Expected Results

Upon completion, the system is expected to demonstrate the following outcomes:

1. **Pronunciation Classification:** The SVM classifier successfully categorizes user pronunciation into predefined quality classes such as *Good* and *Poor* based on extracted speech features and similarity measurements.
2. **Reliable Feature Extraction:** The system accurately extracts Mel Frequency Cepstral Coefficients (MFCCs) from both reference and user speech recordings and computes meaningful similarity measures using Dynamic Time Warping (DTW).
3. **Pronunciation Scoring:** The system generates an overall pronunciation score that reflects the similarity between the user's pronunciation and the reference pronunciation.
4. **Efficient Evaluation:** The complete pronunciation assessment process, from audio recording to result generation, is expected to be completed within a few seconds on standard computer hardware.
5. **Concrete Execution Sample:**
 - **Input Sentence:** Example Nepali sentence
 - **Reference Audio:** Standard Nepali pronunciation recording
 - **DTW Similarity Score:** Computed from reference and user speech
 - **Pronunciation Score:** 82 / 100
 - **Classification Result:** Good
6. **User Interface:** A functional Streamlit-based application enables users to select or view Nepali sentences, record their speech, and receive an overall pronunciation score along with a pronunciation quality classification.
7. **Automated Pronunciation Evaluation:** The system provides a simple and automated method for assessing Nepali pronunciation quality without requiring manual evaluation by language experts.

Limitation: The proposed system performs only sentence-level pronunciation evaluation. It provides an overall pronunciation score and quality classification but does not identify specific mispronounced phonemes, words, or articulatory errors.

Chapter 6

Appendix

6.1 System Architecture Overview

The proposed AI-based Nepali pronunciation evaluation system follows a speech processing pipeline consisting of data collection, preprocessing, feature extraction, similarity measurement, and classification.

6.2 Workflow Pseudocode

```
def pronunciation_evaluation_system():  
  
    # Step 1: User Input  
    sentence = display_nepali_sentence()  
    audio = record_audio(sample_rate=16000)  
  
    # Step 2: Preprocessing  
    audio = preprocess(audio)  
  
    # Step 3: Feature Extraction  
    mfcc_features = extract_mfcc(audio)  
  
    # Step 4: Reference Comparison  
    reference_mfcc = load_reference(sentence)  
    dtw_distance = compute_dtw(mfcc_features, reference_mfcc  
    )  
  
    # Step 5: Feature Vector  
    feature_vector = combine(mfcc_features, dtw_distance)  
  
    # Step 6: Classification  
    model = load_svm_rbf_model()  
    result = model.predict(feature_vector)  
  
    # Step 7: Output  
    display_result(result)
```

Listing 6.1: Pronunciation Evaluation Pipeline

6.3 Phoneme Reference Note

The system is based on acoustic similarity rather than phoneme-level segmentation. Therefore, it does not explicitly map speech into phonemes but instead evaluates pronunciation quality using MFCC and DTW-based features.

6.4 System Limitation Summary

The system is limited to sentence-level pronunciation evaluation and does not provide phoneme-level error detection, articulatory feedback, or stress/intonation correction. It is designed for overall pronunciation quality classification only.

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